CANALRUNG · TRACEABILITY MATRIX

CanalRung traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/03b-traceability.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-05	Amish Chadha	Generated by .kit/trace.py
0.2	2026-11-05	Amish Chadha	Generated by .kit/trace.py
0.1	2026-11-05	Amish Chadha	Generated by .kit/trace.py

CanalRung traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 13 user requirements, 13 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Reach the water from the bank	CNR-CAL-001	Calculation [A1] to [A5]; deploy on a test wall at TRL 4	Met on paper: 4,020 mm; two rungs in the water on the design slope and four on a 3.0 m vertical wall; works on 1.5:1 slopes to 2.1 m and vertical walls to 3.8 m
UR-007	PR-002	Throwable by one adult	CNR-CAL-001, CNR-BLD-001, CNR-STM-001	Weigh; throw trials with volunteers (TRL 4)	Met on paper on mass: 4.27 kg [B3], 0.23 kg inside; throw accuracy needs trials
UR-008	PR-003	Rungs float at the water line	CNR-CAL-001, CNR-BLD-001	Calculation [C1] to [C3]; tank test (TRL 4)	Met on paper: 0.27 kg (plain) and 0.22 kg (stand-off) per rung; factor 1.75
UR-004	PR-004	Carry a climbing adult	CNR-CAL-001, CNR-BLD-001, CNR-STM-001	Calculation [D1], [E1]; proof-load test (TRL 4)	Met on paper: rung at 0.56 of yield; 9 mm ropes 18.4 kN, 4.08 times 4.5 kN
UR-003	PR-005	Climbable on a flat wall	CNR-CAL-001, CNR-BLD-001, CNR-STM-001	Geometry [A6]; supervised tank trials with lifeguards (TRL 4)	Geometry met by design (298 mm, 335 mm, 70 mm); climb time needs trials
UR-005	PR-006	Anchor holds	CNR-CAL-001, CNR-BLD-001, CNR-STM-001	Calculation [G1] to [G6]; pull test on site (TRL 4)	Met on paper on level ground and in line: eye nut 1.53; footing 3.58 drained, 1.59 saturated; the slope-crest reduction of the footing and the eye nut pull off its axis are not yet checked (CNR-STM-001 S4, S5, S23)
UR-002	PR-007	Deploy quickly	CNR-CAL-001, CNR-BLD-001	Timed trials with untrained volunteers (TRL 4)	Not verifiable at TRL 3; estimate 15 s [H1]
UR-009	PR-008	Survive outdoor storage	CNR-CAL-001, CNR-BLD-001, CNR-STM-001	Accelerated UV ageing and pull test (TRL 4)	Met by design (stored in an opaque UV-stabilised box) [H2]; to confirm by test

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-010	PR-009	Value for money	CNR-CAL-001	Costed bill of materials [I1]	Over target: estimated USD 605.20, USD 105.20 over
UR-006	PR-010	Nothing thrown can hurt the person in the water	CNR-CAL-001	Inspection of the model and BOM [H3]	Met by design
UR-011	PR-011	Built and mended locally	CNR-CAL-001	Inspection of the BOM, the model and the build plan BLD-001	Met by design
UR-002, UR-012	PR-012	Obvious and inspectable station	CNR-CAL-001	Inspection of the model and BOM lines 8 and 11	Met by design
UR-013	PR-013	Honest labelling	CNR-CAL-001	Inspection of the repository; `python .kit/render.py --check`	Met by design

Gaps

None.